



Faculty of Engineering and Computer Science
Department of Electrical and Computer Engineering

PROJECT MANAGEMENT

ELEC/COEN 390 - BB

Electrical/Computer Engineering Product Design Project

Stakeholders

Dr. Wahab Hamou-Lhadj

Team 6

Bilal Samee
27286295

Dominique Reynolds-Sandy
40241168

Navraj Jhajj
40129282

Armaan Khan
40235610

Dipita Sinha
40273009

**We certify that this submission is our original work and meets the
Faculty's Expectations of Originality**

August 15, 2026



Automated Aquarium Monitoring Systems (AAMS Inc.)

AQUASENSE

Project Management • Rev. 1.0.0

Contents

List of Figures	ii
List of Tables	iii
1. Introduction	1
2. Standards	1
2.1. Jira	1
2.2. Confluence	2
2.3. Gitlab	3
2.3.1. CI/CD Pipeline	5
2.4. Discord	10
2.5. Typst	10
3. Standup Meeting Minutes	10

List of Figures

Figure 1	Example Jira issue created during the early stages of the project.	1
Figure 2	Example meeting minutes documented on Confluence.	2
Figure 3	Feature branches resolved, approved, and merged to-date.	3
Figure 4	A chain of thorough code review threads; discussed and addressed.	4
Figure 5	Example of the level of detail and pushback involved in the code reviews.	5
Figure 6	History of pipelines that have ran in recent merges.	6
Figure 7	Gitlab CI/CD pipeline visualization flow diagram.	7
Figure 8	Example of the kind of brainstorming members regularly engaged in. .	10

List of Tables

Table 1	Meeting minutes of the standup meeting on 03/07/2026.	11
Table 2	Meeting minutes of the standup meeting on 06/07/2026.	11
Table 3	Meeting minutes of the standup meeting on 10/07/2026.	12
Table 4	Meeting minutes of the standup meeting on 13/07/2026.	13
Table 5	Meeting minutes of the standup meeting on 17/07/2026.	14
Table 6	Meeting minutes of the standup meeting on 20/07/2026.	14
Table 7	Meeting minutes of the standup meeting on 24/07/2026.	15
Table 8	Meeting minutes of the standup meeting on 27/07/2026.	17
Table 9	Meeting minutes of the standup meeting on 31/07/2026.	17
Table 10	Meeting minutes of the standup meeting on 03/08/2026.	18
Table 11	Meeting minutes of the standup meeting on 05/08/2026.	19
Table 12	Meeting minutes of the standup meeting on 07/08/2026.	20

1. Introduction

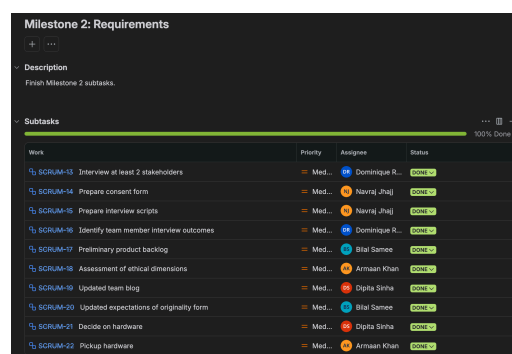
Standards were established from day one, with the aim of assuring quality, encouraging retrospective feedback, and acquainting the less familiar members with industry standards. This document outlines the process, tools, and standards established and maintained by the team throughout the development cycle. It also documents the meeting minutes of all standup meetings conducted, which occurred twice a week.

2. Standards

This section outlines the process, tools, and standards established and maintained by the team throughout the development cycle. The aim is to document the process and its correspondence to quality assurance. The majority of the tools were selected due to their prevalent use in industry, not following from a thorough analysis of tradeoffs.

2.1. Jira

For project management, the team chose Jira. Project management tasks included creating issues with proper descriptions, assigning them to team members, changing their status as work progressed, and categorizing them by user story.



Work	Priority	Assignee	Status
↳ SCRUM-13 Interview at least 2 stakeholders	Med...	Dominique R...	Complete
↳ SCRUM-14 Prepare consent form	Med...	Naveej Jhaj	Complete
↳ SCRUM-15 Prepare interview scripts	Med...	Naveej Jhaj	Complete
↳ SCRUM-16 Identify team member interview outcomes	Med...	Dominique R...	Complete
↳ SCRUM-17 Preliminary product backlog	Med...	Bilal Samee	Complete
↳ SCRUM-18 Assessment of ethical dimensions	Med...	Armaan Khan	Complete
↳ SCRUM-19 Updated team blog	Med...	Digita Sinha	Complete
↳ SCRUM-20 Updated expectations of originality form	Med...	Bilal Samee	Complete
↳ SCRUM-21 Decide on hardware	Med...	Digita Sinha	Complete
↳ SCRUM-22 Pickup hardware	Med...	Armaan Khan	Complete

Figure 1: Example Jira issue created during the early stages of the project.

2.2. Confluence

For process documentation, the team chose Confluence. The only form of process documentation that took place was meeting minutes pertaining to the bi-weekly standups that took place on Monday and Friday. Meeting minutes followed the standard 3-talking-point format of standups, followed by a brief announcements section for anything that needed to be known by the team for the coming week, such as a member being gone for some time or a deadline coming up.

[24 JUL 2026 @ 7:00-7:30] Meeting Minutes: Standup

By Bilal Samee 1 min See views Add a reaction

Team Member	What did you do since last standup?	What are you going to do till next standup?	Blockers	Attendance
Bilal	- Worked on pipeline - Working on FB-1.1 for schema and pytest unit tests and CI/CD pipeline	- Finish DASH-02f - Revamp DASH design in other issue - Work on DASH-03 - Review MRed branches - Work on report	No blockers	YES
Dominique	- Finished HW code for Arduino - Reconfigured circuit - Addressed some MR reviews - Finished troubleshooting page	- Finish addressing MR issues - Work on SR-4.1, ST-5.1 - Work on report	Temperature sensor not working atm	YES
Armaan	- Finished NF-1.1 and NF-1.2 - Addressed MR reviews	- Work on NF-2.1 and NF-2.2 - Work on report	No blockers	YES
Navraj	Finished DASH-4.1	- Brainstorm advanced settings options - Work on NF-03	DASH-4.2/4.3/4.4 blocked by SETTINGS-03 and SETTINGS-04	YES
Dipita	- Finished firebase auth, registration and login into MRs - Firebase schema and rules consultation, proposal, and checkup - Working on SETTINGS-03	Finishing SETTINGS-03 and SETTINGS-04	Blocked by DASH-02f	YES

- REVIEW, REVIEW, REVIEW.
- WE HAVE SPRINT 3 ASSIGNMENT DUE ON THE 26TH.
- EVERYONE SHOULD WORK ON EVERY PART THEY CAN CONTRIBUTE TO OF THE REPORT, ESPECIALLY PERTAINING TO THEIR CODE.
- Bilal will hopefully be back by the night of 27 July.
- Navraj will be out of office Saturday afternoon.
- Dipita will be out of office Saturday evening.

Figure 2: Example meeting minutes documented on Confluence.

2.3. Gitlab

For source version control, the team chose Gitlab. The main branch was set to protected, essentially disabling direct commits being pushed to main. The team enforced feature branches, which were properly scoped, brief, and created specifically to address one and only one feature each. Work was broken down into commits. After work on a branch was done, its merge request was set for review and at least 2 team members, one of which was experienced, needed to review and approve the code in order for it to be merged.

Fb refactor user account I25 · created 1 day ago by b-samee	Merged ✓	Approved	Resolved	updated 21 hours ago
CICD-3.1 Fix Missing Condition I21 · created 2 days ago by b-samee	Merged	Approved		updated 2 days ago
Added deliverable files in docs I17 · created 3 days ago by Dipita Sinha	Merged	Approved	Resolved	updated 2 days ago
CICD-03 hardware compilation I16 · created 4 days ago by b-samee	Merged	Approved	Resolved	updated 2 days ago
Full hardware sensor arduino c code I15 · created 5 days ago by Dominique Reynolds-Sandy	Merged ✓	Approved	Resolved	updated 20 hours ago
Fb 1.1p2 firebase bringup I14 · created 5 days ago by b-samee	Merged	Approved	Resolved	updated 2 days ago
Firebase Setup I13 · created 6 days ago by Dipita Sinha	Merged	Approved	Resolved	updated 5 days ago
Dash 02f dashboard redesign I11 · created 1 week ago by b-samee	Merged	Approved	Resolved	updated 1 day ago
Fb 1.1p1 firebase schema I10 · created 1 week ago by b-samee	Merged	Approved	Resolved	updated 5 days ago
Settings 02 finalize settings & contact I6 · created 1 week ago by Dipita Sinha	Merged	Approved	Resolved	updated 1 week ago
[DASH-02e] Refactored aquarium sensors I5 · created 1 week ago by b-samee	Merged	Approved	Resolved	updated 1 week ago
[SETTINGS-01] Implemented Base Settings Page UI I3 · created 2 weeks ago by b-samee	Merged	Approved	Resolved	updated 1 week ago
[DASH-01] Main dashboard elements I2 · created 2 weeks ago by b-samee	Merged	Approved	Resolved	updated 1 week ago
Scrum 23 scopedlogger sharedpreferencehelper I1 · created 1 month ago by b-samee	Merged	Approved		updated 3 weeks ago

Figure 3: Feature branches resolved, approved, and merged to-date.

Code reviews were thorough and pushback was encouraged, albeit with the intention of satisfying the reviewer's comments as best as feasible. After each review was addressed, the reviewer themselves would double-check the solution and resolve the review thread. With all threads resolved, they would approve the merge request. After two approvals, as long as the branch is in-sync with main and passes the CI/CD pipeline, it is merged.

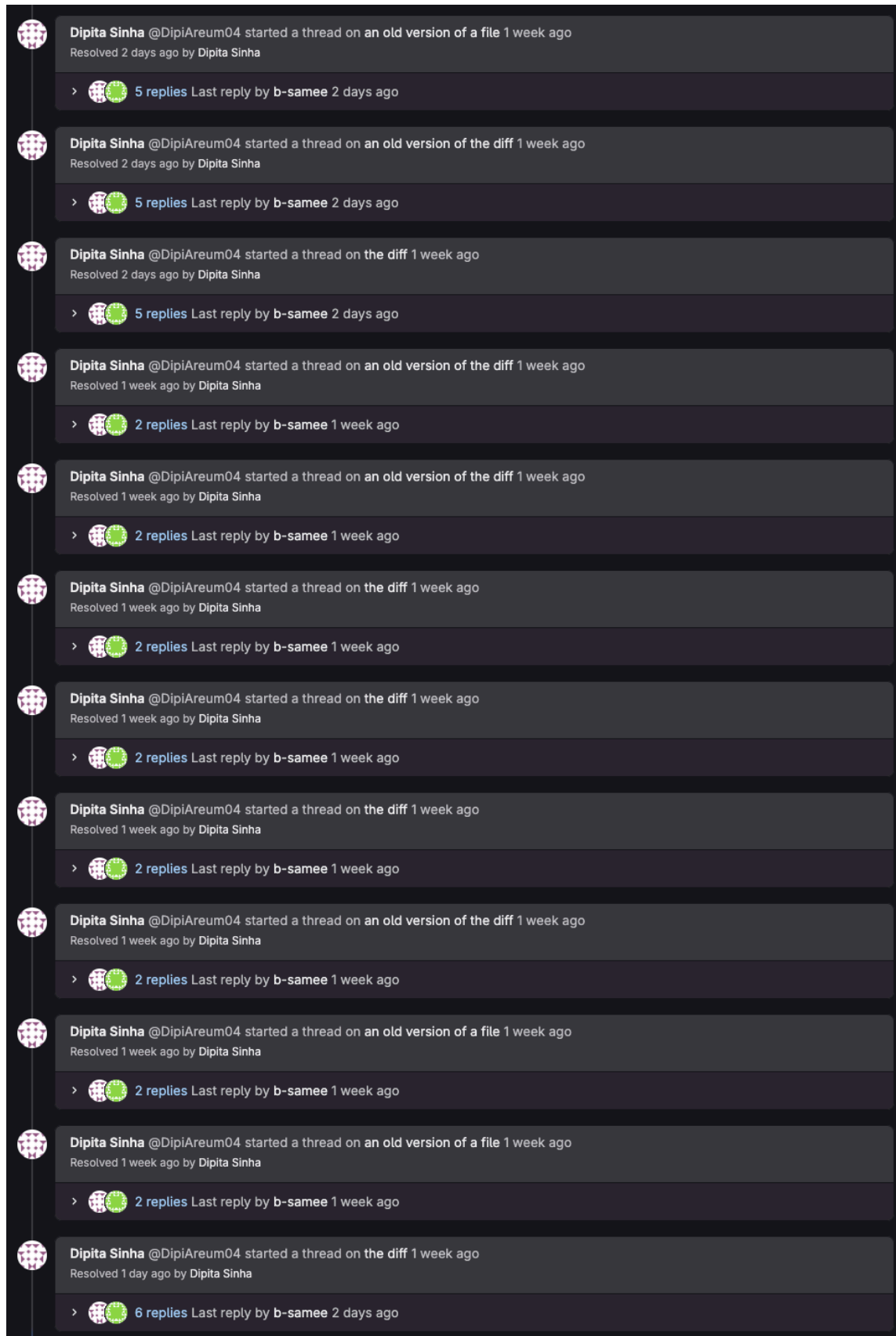


Figure 4: A chain of thorough code review threads; discussed and addressed.

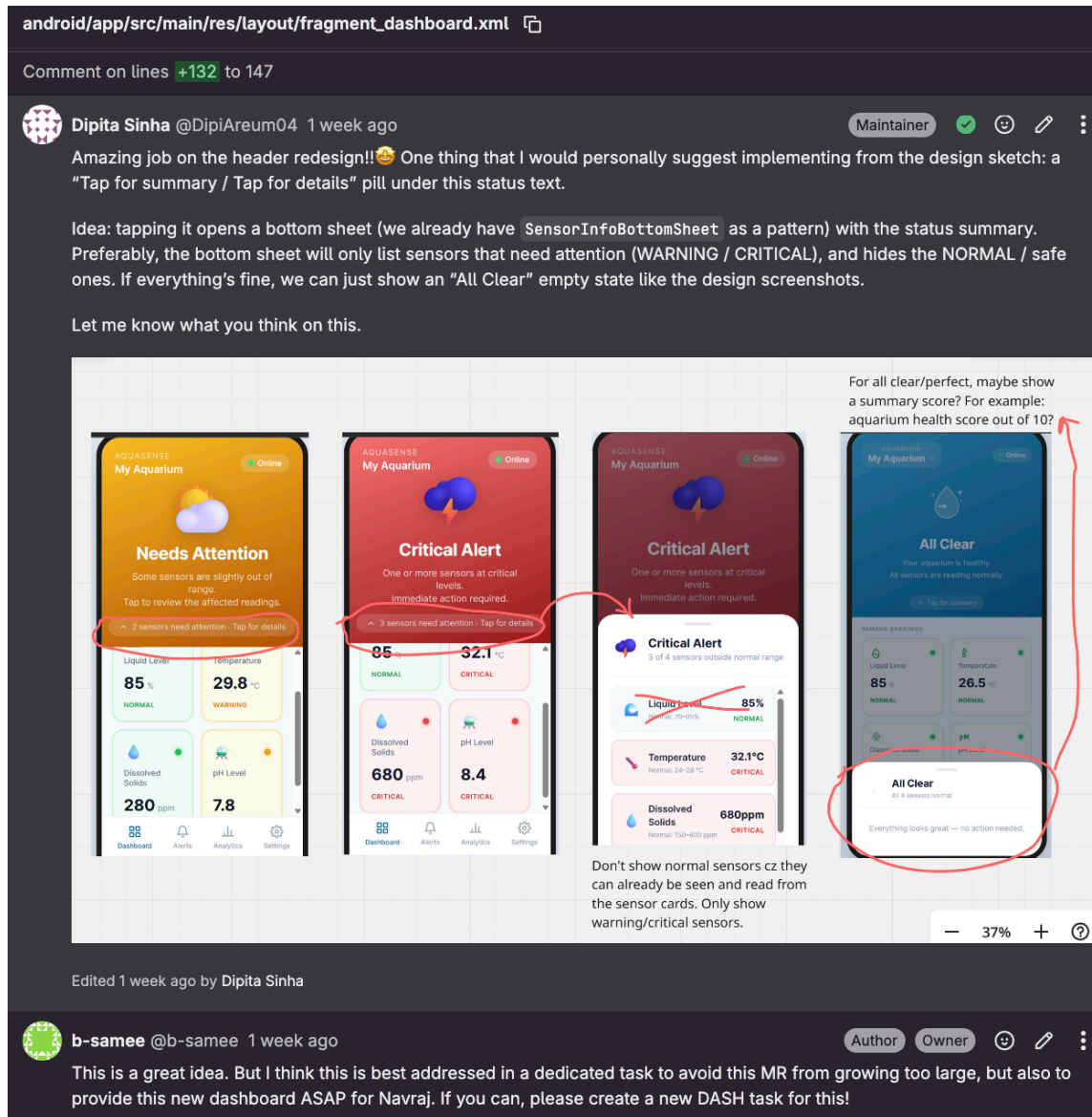


Figure 5: Example of the level of detail and pushback involved in the code reviews.

2.3.1. CI/CD Pipeline

Over the course of previous sprints, the team has developed a CI/CD pipeline using Gitlab's shared runners. Figure 7 outlines the architecture and dependency flow of the CI/CD pipeline. For source code, a setup stage prepares environment variables and installs required dependencies. This is followed by a build stage, where compilation is done. Finally, a lint stage performs linting and static checks. For the database system itself, a test stage executes unit

tests on a development database against the database schema as well as a validate stage that safely executes validation tests against the production database.

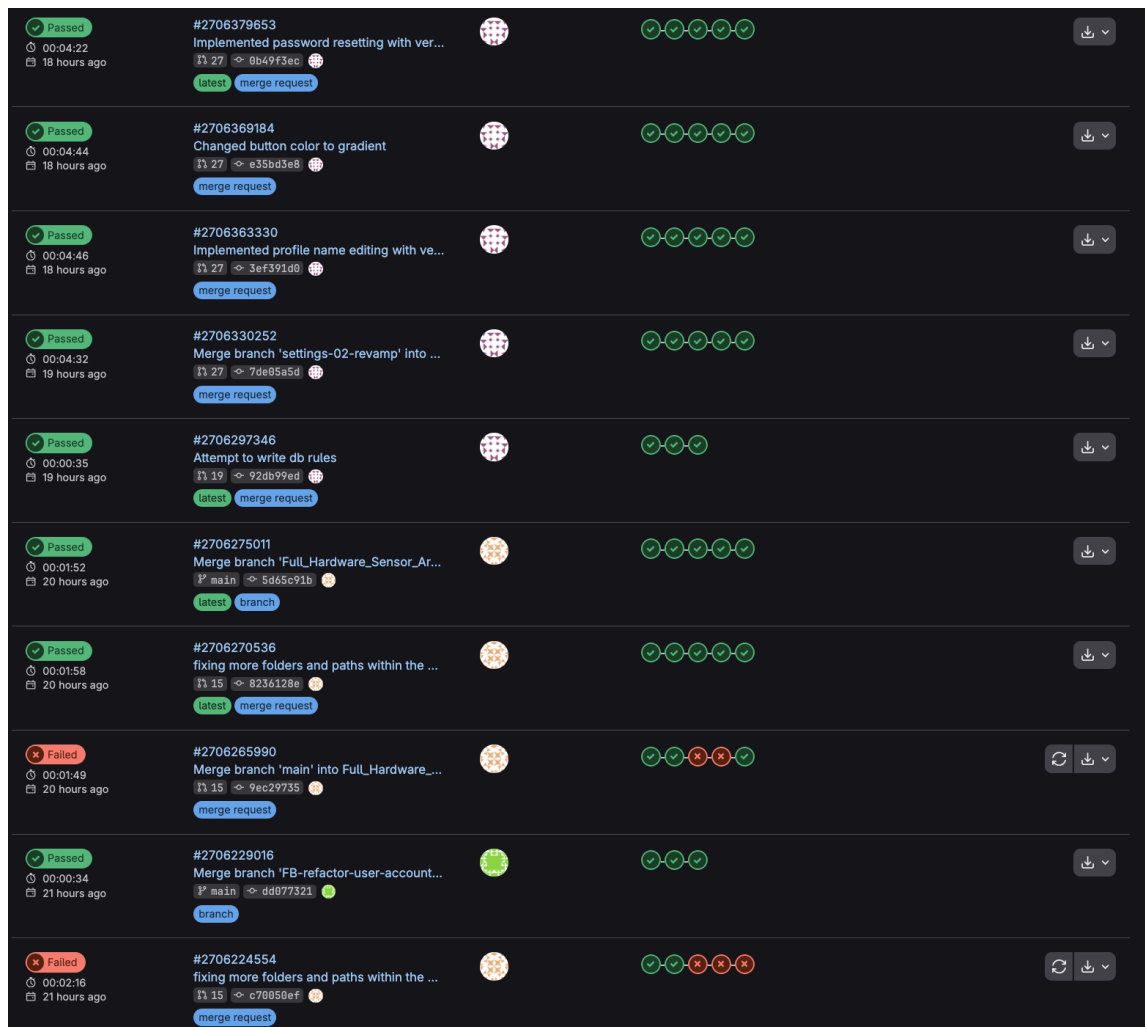


Figure 6: History of pipelines that have ran in recent merges.

All tests are written in Python using the pytest framework, with proper use of mocks, fixtures, and test functions. Credentials are supplied to pipeline stages using proper secrets and environment variables, not hard-coded into files that would be pushed to the repository.

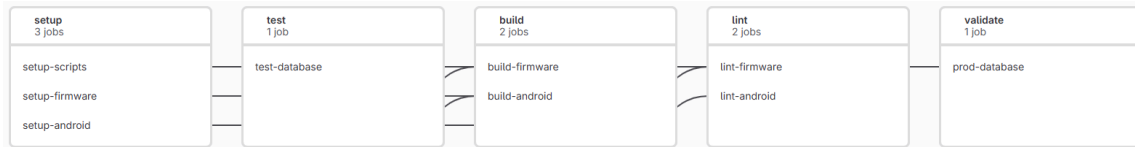


Figure 7: Gitlab CI/CD pipeline visualization flow diagram.

Snippet 1 details the pipeline stages and their respective jobs. To save on resources, the pipeline itself only runs on merge requests then on the main branch if any merge was performed. Moreover, pipeline jobs are tied to specific folder contents being modified.

Snippet 1: .gitlab-ci.yml

```

1  # A pipeline is composed of independent jobs that run scripts, grouped into stages.
2  # Stages run in sequential order, but jobs within stages run in parallel.
3  # For more information, see: https://docs.gitlab.com/ee/ci/yaml/#stages.
4  image: ghcr.io/astral-sh/uv:python3.14-bookworm-slim
5
6  workflow:
7    rules:
8      - if: '$CI_PIPELINE_SOURCE == "merge_request_event"'
9      - if: '$CI_COMMIT_BRANCH == "main" && $CI_PIPELINE_SOURCE == "push"'
10
11  stages:
12    - setup
13    - build
14    - lint
15    - validate
16
17  setup-scripts:
18    stage: setup
19    script:
20      - cd scripts && uv sync
21    artifacts:
22      expire_in: 1 hour
23      paths:
24        - scripts/.venv
25
26  setup-firmware:
27    image: ubuntu:22.04
28    stage: setup
29    rules:
30      - changes:
31        - firmware/**/*
32
33  variables:
34    ARDUINO_CLI_VERSION: "1.5.1"
35    ARDUINO_DIRECTORIES_DATA: "${CI_PROJECT_DIR}/.arduino15"
36    ARDUINO_DIRECTORIES_DOWNLOADS: "${CI_PROJECT_DIR}/.arduino-downloads"
37    ARDUINO_DIRECTORIES_USER: "${CI_PROJECT_DIR}/Arduino"
38  script:
39    - apt-get update && apt-get install -y curl
40    - mkdir -p "${CI_PROJECT_DIR}/bin"
41    - curl -fsSL https://raw.githubusercontent.com/arduino/arduino-cli/master/install.sh |
42      BINDIR="${CI_PROJECT_DIR}/bin" sh -s v${ARDUINO_CLI_VERSION}
43    - export PATH="${CI_PROJECT_DIR}/bin:$PATH"
44    - arduino-cli config init
45    - arduino-cli core update-index
46    - arduino-cli core install arduino:renesas_uno
47    - arduino-cli lib update-index
48    - arduino-cli lib install "OneWire"
49    - arduino-cli lib install "DallasTemperature"
50    - arduino-cli lib install "ArduinoHttpClient"
51    - arduino-cli lib install "ArduinoJson"

```

```

52   artifacts:
53     expire_in: 1 hour
54     paths:
55       - bin/
56       - .arduino15/
57       - .arduino-downloads/
58       - Arduino/
59
60   setup-android:
61     image: eclipse-temurin:17-jdk
62     stage: setup
63     rules:
64       - changes:
65         - android/**/*
66     variables:
67       ANDROID_COMPILE_SDK: "36"
68       ANDROID_BUILD_TOOLS: "36.0.0"
69       ANDROID_SDK_TOOLS: "11076708"
70     script:
71       - cd android
72       - apt-get --quiet update --yes
73       - apt-get --quiet install --yes wget unzip
74       - cp "$GOOGLE_SERVICES_JSON" app/google-services.json
75       - export ANDROID_HOME="${PWD}/android-sdk-root"
76       - install -d $ANDROID_HOME
77       - wget --quiet --output-document=$ANDROID_HOME/cmdline-tools.zip "https://dl.google.com/android/
repository/commandlinetools-linux-${ANDROID_SDK_TOOLS}_latest.zip"
78       - unzip -q -d "$ANDROID_HOME/cmdline-tools" "$ANDROID_HOME/cmdline-tools.zip"
79       - mv $ANDROID_HOME/cmdline-tools/cmdline-tools $ANDROID_HOME/cmdline-tools/latest
80       - export PATH="$PATH:$ANDROID_HOME/cmdline-tools/latest/bin:$ANDROID_HOME/platform-tools"
81       - yes | sdkmanager --sdk_root=$ANDROID_HOME --licenses > /dev/null || true
82       - sdkmanager --sdk_root=$ANDROID_HOME "platforms;android-${ANDROID_COMPILE_SDK}" "build-tools;
${ANDROID_BUILD_TOOLS}"
83       - chmod +x ./gradlew
84     artifacts:
85       expire_in: 1 hour
86       paths:
87         - android/android-sdk-root/
88
89   build-firmware:
90     image: ubuntu:22.04
91     stage: build
92     rules:
93       - changes:
94         - firmware/**/*
95     needs:
96       - job: setup-firmware
97     artifacts: true
98     variables:
99       FQBN: "arduino:renesas_uno:unor4wifi"
100      ARDUINO_DIRECTORIES_DATA: "${CI_PROJECT_DIR}/.arduino15"
101      ARDUINO_DIRECTORIES_DOWNLOADS: "${CI_PROJECT_DIR}/.arduino-downloads"
102      ARDUINO_DIRECTORIES_USER: "${CI_PROJECT_DIR}/Arduino"
103    before_script:
104      - cd firmware/src
105      - f=(*.ino); mv -n "$f" "$(basename "$PWD").ino"
106      - export PATH="${CI_PROJECT_DIR}/bin:$PATH"
107    script:
108      - arduino-cli compile --fqbn $FQBN .
109
110   build-android:
111     image: eclipse-temurin:17-jdk
112     stage: build
113     rules:
114       - changes:
115         - android/**/*
116     needs:
117       - job: setup-android
118     artifacts: true
119     before_script:
120       - cd android
121       - cp "$GOOGLE_SERVICES_JSON" app/google-services.json
122       - export ANDROID_HOME="${PWD}/android-sdk-root"
123       - export PATH="$PATH:$ANDROID_HOME/cmdline-tools/latest/bin:$ANDROID_HOME/platform-tools"
124       - chmod +x ./gradlew
125     script:
126       - ./gradlew assembleDebug --stacktrace
127     artifacts:
128       expire_in: 1 hour
129     paths:

```

```

131     - "**/build/outputs/apk/debug/*.apk"
132     when: on_success
133     cache:
134       key: ${CI_PROJECT_ID}
135       paths:
136         - .gradle/
137
138
139     lint-firmware:
140       image: ubuntu:22.04
141       stage: lint
142       rules:
143         - changes:
144           - firmware/**/*
145       needs:
146         - job: setup-firmware
147         artifacts: true
148       variables:
149         FQBN: "arduino:renesas_uno:unor4wifi"
150         ARDUINO_DIRECTORIES_DATA: "${CI_PROJECT_DIR}/.arduino15"
151         ARDUINO_DIRECTORIES_DOWNLOADS: "${CI_PROJECT_DIR}/.arduino-downloads"
152         ARDUINO_DIRECTORIES_USER: "${CI_PROJECT_DIR}/Arduino"
153       before_script:
154         - cd firmware/src
155         - apt-get update && apt-get install -y clang-tidy jq
156         - export PATH="${CI_PROJECT_DIR}/bin:$PATH"
157       script:
158         - arduino-cli compile --fqbn $FQBN --only-compilation-database --build-path build .
159         - GXX_BIN=$(find "${CI_PROJECT_DIR}/.arduino15/packages/arduino/tools/arm-none-eabi-gcc" -name "arm-
160 none-eabi-g++" | head -n1)
161         - echo "Using compiler at $GXX_BIN"
162         - INCLUDE_DIRS=$(("$GXX_BIN" -E -x c++ - -v < /dev/null 2>&1 | sed -n '/#include <...> search starts
163 here:./,/End of search list/p' | sed 'ld;$d' | sed 's/^ *//'))
164         - echo "Discovered include dirs:"
165         - echo "$INCLUDE_DIRS"
166         - EXTRA_ARGS=""
167         - for dir in $INCLUDE_DIRS; do EXTRA_ARGS="$EXTRA_ARGS --extra-arg=-isystem$dir"; done
168         - jq -r '.[].file' build/compile_commands.json | grep "$(pwd)" | sort -u > files_to_lint.txt
169         - xargs -r clang-tidy -p build --extra-arg--target=arm-none-eabi $EXTRA_ARGS < files_to_lint.txt
170
171     lint-android:
172       image: eclipse-temurin:17-jdk
173       stage: lint
174       rules:
175         - changes:
176           - android/**/*
177       needs:
178         - job: setup-android
179         artifacts: true
180       before_script:
181         - cd android
182         - cp "$GOOGLE_SERVICES_JSON" app/google-services.json
183         - export ANDROID_HOME="${PWD}/android-sdk-root"
184         - export PATH="$PATH:$ANDROID_HOME/cmdline-tools/latest/bin:$ANDROID_HOME/platform-tools"
185         - chmod +x ./gradlew
186       script:
187         - ./gradlew lintDebug --stacktrace
188       artifacts:
189         expire_in: 1 hour
190         when: always
191         paths:
192           - "**/build/reports/lint-results-debug.html"
193           - "**/build/reports/lint-results-debug.xml"
194       cache:
195         key: ${CI_PROJECT_ID}
196         paths:
197           - .gradle/
198
199     prod-database:
200       stage: validate
201       needs:
202         - job: setup-scripts
203         artifacts: true
204       script:
205         - cd scripts
206         - cp "$PROD_APP_CONFIG_JSON" ./ProdDbAppConfig.json
207         - cp "$PROD_SERVICE_ACCOUNT_KEY" ./ProdDbServiceAccountKey.json
208         - source .venv/bin/activate
209         - pytest -vv tests/test_production_database.py

```

2.4. Discord

For team communications, the team chose Discord. Despite Microsoft Teams or Slack arguably being more prevalent, it was more convenient to use Discord due to the team's familiarity and availability on the platform.

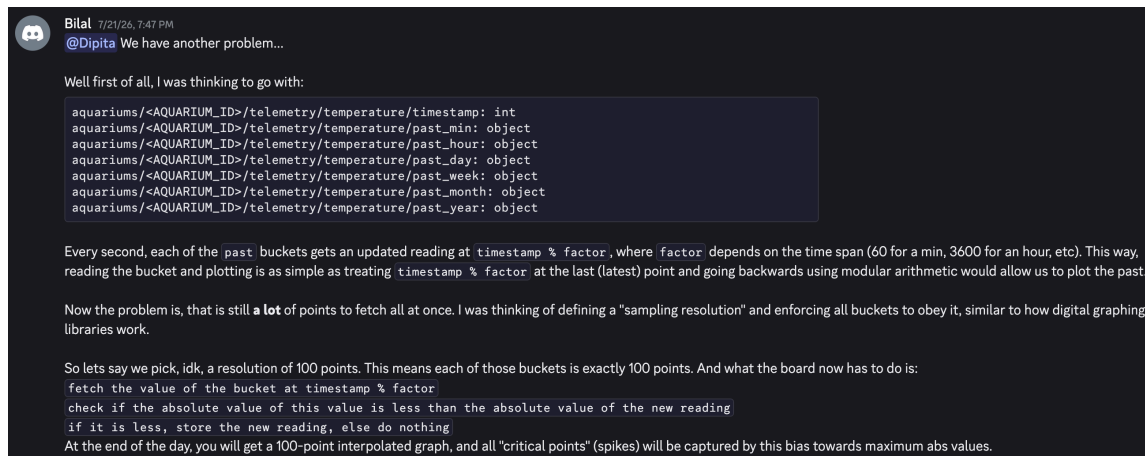


Figure 8: Example of the kind of brainstorming members regularly engaged in.

2.5. Typst

For report typesetting, the team chose Typst. Despite LaTeX being the industry-standard for typesetting, it was more efficient to make use of Typst due to its similarity to Markdown and scripting languages, which the majority of the team was familiar with. The use of a typesetting system allowed the team to standardize the creation of reports efficiently and with minimal formatting inconsistencies.

3. Standup Meeting Minutes

Appendix A tabulates the meeting minutes of all standup meetings conducted throughout sprints 1 to 3. Standups were conducted twice a week, on Monday and Friday, officially from 7:00 PM to 7:30 PM, using the standard standup 3-question format. Topics discussed during standups needed to stay within the scope of the standup and anything else that needed to

be discussed was done after the standup meeting concluded or as part of other meetings, not recorded in this document.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
Bilal	Nothing.	Nothing.	No blockers.	✓
Dominique	Worked on privacy policy page; completed research, currently writing draft.	Will finish privacy page by Monday/ Tuesday.	No blockers.	✓
Armaan	Measured components for final design. Wired up circuits to study connectivity.	Plan to simulate the circuit to analyze power. Programming as per SR1.1.	No blockers.	✓
Navraj	Nothing.	Nothing.	No blockers.	✓
Dipita	Brainstormed Settings UI designs and finalized Settings layout; researched sensor calibration methods and wrote draft for calibration guides.	Nothing.	No blockers.	✓

Table 1: Meeting minutes of the standup meeting on 03/07/2026.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
-------------	-------------------------------------	-------------------------------------	--	------------------

Bilal	Nothing.	Dashboard tasks.	No blockers.	✓
Dominique	Still working on privacy page.	Finishing Privacy by tmr night & work on SR-2.1 & SR-2.2.	No blockers.	✓
Armaan	KiCAD circuit.	Plan to simulate the circuit to analyze power. Programming as per SR1.1.	No blockers.	✓
Navraj	Nothing.	Sensor spacing and sensor measurement integrity, cable management, etc.	No blockers.	✓
Dipita	Nothing.	Will finish implementing Settings page by Friday.	No blockers.	✓

Table 2: Meeting minutes of the standup meeting on 06/07/2026.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
Bilal	Worked on dashboard features.	Finish up dashboard features. Work on sprint 2 report.	No blockers.	×
Dominique	Finished the privacy page. Began backbone code for sensor reading.	Integrate to the internet. Work on Firebase. Add Firebase to the privacy policy.	No blockers.	✓

Armaan	Simulated the circuit.	Start my portion of the software.	No blockers.	✓
Navraj	Planned the 3D shelf design taking in measurements and distance. Created a CAD design.	Work on Firebase.	No blockers.	✓
Dipita	Implemented the base settings UI.	Integrate with the main dashboard. Finalize the setting page. Work on the contact page.	No blockers.	✓

Table 3: Meeting minutes of the standup meeting on 10/07/2026.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
Bilal	Worked on DASH-01, DASH-02.	Merge DASH-01. Start working on DASH-03.	No blockers.	✓
Dominique	Worked on SENSOR-02, SENSOR-03, PRIVSEC-01, page designs.	Work on SENSOR-02, SENSOR-04, SETTINGS-05.	No blockers.	✓
Armaan	Worked on SR-4.2.	Work on SR-4.1, NF-1.1/1.2.	No blockers.	✓
Navraj	Worked on 3D model fix, helped with firebase.	Work on DASH-04.	No blockers.	✓
Dipita	Worked on SETTINGS-01, SETTINGS-02.	Merge SETTINGS-01. Try to finish	No blockers.	✓

		SETTINGS-03. Make aesthetics task.		
--	--	--	--	--

Table 4: Meeting minutes of the standup meeting on 13/07/2026.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
Bilal	Merged DASH-01. Code reviews. Merge DASH-02e.	Work on DASH-03.	No blockers.	✓
Dominique	Finished SENSOR-02.	Work on SENSOR-04. Combining all sensor code together.	No blockers.	✓
Armaan	Pushed TDS and WL sensor code.	Work on NF-1.1/1.2. Test personal Arduino until we get new board.	No blockers.	✓
Navraj	Worked on DASH-04.	Finish DASH-04.	No blockers.	✓
Dipita	Updated product backlog. Started SETTINGS-03. Finished SETTINGS-02 merge.	Finishing SETTINGS-03.	No blockers.	✓

Table 5: Meeting minutes of the standup meeting on 17/07/2026.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
-------------	-------------------------------------	-------------------------------------	--	------------------

Bilal	Working on a new DASH-02f to revamp design. Working on FB-1.1 for schema and pytest unit tests and CI/CD pipeline.	Working on a new DASH-02f to revamp design. Working on FB-1.1 for schema and pytest unit tests and CI/CD pipeline. Work on DASH-03.	No blockers.	✓
Dominique	Finished SR-2.4, FB-1.2, HW code for Arduino.	Working on ST-5.1, ST-5.2. Working on SR-4.1, SR-4.2.	No blockers.	✓
Armaan	Rewired circuit using personal microcontroller. Working on NF-1.2.	NF-1.1/1.2 and the others. Test personal Arduino tonight.	No blockers.	✓
Navraj	Collected tentative stakeholder opinion.	Addressing merge requests. Finish DASH-04.	No blockers.	✓
Dipita	Worked on FB-2.1. Worked on SETTINGS-03. Worked on DASH designs and other UI stuff.	Finishing SETTINGS-03. Finishing FB-02. Start SETTINGS-04.	No blockers.	✓

Table 6: Meeting minutes of the standup meeting on 20/07/2026.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
Bilal	Worked on pipeline. Working on	Finish DASH-02f. Revamp DASH	No blockers.	✓

	FB-1.1 for schema and pytest unit tests and CI/CD pipeline.	design in other issue. Work on DASH-03. Review MRed branches. Work on report.		
Dominique	Finished HW code for Arduino. Reconfigured circuit. Addressed some MR reviews. Finished troubleshooting page.	Finish addressing MR issues. Work on SR-4.1, ST-5.1. Work on report.	Temperature sensor not working atm.	✓
Armaan	Finished NF-1.1 and NF-1.2. Addressed MR reviews.	Work on NF-2.1 and NF-2.2. Work on report.	No blockers.	✓
Navraj	Finished DASH-4.1.	Brainstorm advanced settings options. Work on NF-03.	DASH-4.2/4.3/4.4 blocked by SETTINGS-03 and SETTINGS-04.	✓
Dipita	Finished firebase auth, registration and login into MRs. Firebase schema and rules consultation, proposal, and checkup. Working on SETTINGS-03.	Finishing SETTINGS-03 and SETTINGS-04.	Blocked by DASH-02f.	✓

Table 7: Meeting minutes of the standup meeting on 24/07/2026.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
Bilal	Worked on DASH-05.	Finishing up DASH-05. Finishing code review.	No blockers.	✓
Dominique	Finished SR-2.4. Finished ST-5.1. Finished BLE-1.1 and BLE-1.2. Fixed Temperature sensor code.	Finish addressing MR issues. Finishing up ST-5.2.	No blockers.	✓
Armaan	Worked on NF-1.1 and NF-1.2.	Finish addressing MR issues on NF-1.1 and NF-1.2.	No blockers.	✓
Navraj	Worked on DASH-4.1 and 4.4. Worked on NF-3.1 and 3.2.	Finish addressing MR issues on DASH-04.	No blockers.	✓
Dipita	Finished FB-2.1 and 2.2, also FB-1.3. Worked on SETTINGS-03. Worked on SETTINGS-02 revamp.	Finishing SETTINGS-03 and SETTINGS-04. Finishing NF-1.3 UI.	No blockers.	✓

Table 8: Meeting minutes of the standup meeting on 27/07/2026.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
Bilal	Worked on mock aquarium for demoing,	Review Nav and Armaan's first branches plus Dom's	No blockers.	✓

	worked on dash wiring.	current branch, finish dash wiring and revamp tonight, work on Dash 03.		
Dominique	Finished ST-5.2, addressed Dipita MR issues but still working on remaining issues.	Finish MR issues, BLE-02, look at Armaan's code for SENSOR-04.	No blockers.	✓
Armaan	Finished addressing NF-1.1/1.2, finished NF-2.1.	Address any issues mentioned, do wiring for NF-1.3, do NF-2.2/2.3.	IT issue with firebase signin network error.	✓
Navraj	Fixed 3D casing.	Finish DASH-04, work on NOTIF-03, stakeholder interviews, address any comments.	No blockers.	✓
Dipita	Redesigned SETTINGS-04 and did UI, researched threshold values with citations, finished NF-1.3 UI.	Finish SETTINGS-03/04, work on BLE-02.	Need schema updated.	✓

Table 9: Meeting minutes of the standup meeting on 31/07/2026.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
-------------	-------------------------------------	-------------------------------------	--	------------------

Bilal	Finished mock aquarium and DB schema, finished dash wiring, finished HW code review and merge.	Start and finish DASH 05, work on DASH 03.	No blockers.	✓
Dominique	Started work on pH calibration, address MR comments, started BLE-02 backend.	Finish BLE-02 backend tonight, look over MRs, work on pH calibration, design and impl loading screen.	No blockers.	✓
Armaan	Finished NF-2.1/2.2, addressed MR comments.	Do wiring for NF-1.3/2.3, print 3D case.	No blockers.	✓
Navraj	Finished DASH-04, 3D case, finished NF-3.1/3.2.	Work on NF-3.3, work on PRIVSEC-02.	No blockers.	✓
Dipita	Finished BLE-02 frontend, did code reviews, worked on SETTINGS-04, worked on DB rules.	Finish SETTINGS-03/04	No blockers.	✓

Table 10: Meeting minutes of the standup meeting on 03/08/2026.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
Bilal	Working on DASH-05.	Finish DASH 05, work on DASH 03.	No blockers.	✓

Dominique	pH calibration done, BLE-02 done, firmware addressing done.	Work on loading screen.	No blockers.	×
Armaan	Did wiring for NF-1.3, finished NF-2.3.	Print 3D case. Start PRIVSEC-02.	No blockers.	✓
Navraj	Reworked 3D case, started NF-3.3.	Print 3D case. Work on NF-3.3.	No blockers.	✓
Dipita	Finished SETTINGS-04. Started working on DASH-04 revamp branch.	Finish DASH-04 revamp branch, SETTINGS-03.	Waiting on BLE-02 HW branch to test BLE-02 frontend.	✓

Table 11: Meeting minutes of the standup meeting on 05/08/2026.

Team Member	What did you do since last standup?	What will you do till last standup?	Are there any blockers you are facing?	Attended Meeting
Bilal	Work on DASH-05.	Make sure DASH-05 is merged tonight, finish DASH-05 followup for sensor info tooltips, start and finish DASH-03, work on merging everything else.	No blockers.	✓
Dominique	Worked on loading screen, will finish tonight.	Work on PRIVSEC-02.	No blockers.	✓

Armaan	Printed 3D case.	Work on SETTINGS-06.	No blockers.	✓
Navraj	Worked on 3D design, printed 3D case.	Nothing.	No blockers.	×
Dipita	DASH-04 revamp done, refactored BLE-02 frontend, finalized casing design.	Work on and finish SETTINGS-03. Reprint casing.	No blockers.	✓

Table 12: Meeting minutes of the standup meeting on 07/08/2026.